

Module 30.1: Cash Flow Introduction and Direct Method CFO

SCHWESERNOTES - BOOK 2

LOS 30.a: Describe how the cash flow statement is linked to the income statement and the balance sheet.

The **cash flow statement** provides information for a reporting period beyond that available from the income statement, which is based on accrual, rather than cash, accounting. Analysts use cash flow statements to understand:

- A company's cash receipts and cash payments during an accounting period
- A company's operating, investing, and financing activities
- The impact of accrual accounting events on cash flows
- A company's quality of earnings

An analyst can use the statement of cash flows to assess a firm's liquidity, solvency, and financial flexibility, including:

- Whether regular operations generate enough cash to sustain the business
- Whether the firm generates enough cash to pay off existing debts as they mature
- Whether the firm is likely to need additional financing
- Whether the firm can meet unexpected obligations
- Whether the firm can take advantage of new business opportunities as they arise

The cash flow statement reconciles the beginning and ending balances of cash in the balance sheet over an accounting period. The change in cash is a result of the firm's operating, investing, and financing activities, as follows:

	Operating cash flow
+/-	Investing cash flow
+/-	Financing cash flow
=	Change in cash balance
+	Beginning cash balance
=	Ending cash balance

Earnings are considered to be of high quality when operating cash flows, which can also be called cash flow from operations (CFO), are close to or higher than reported earnings. If earnings are consistently higher than CFO, their quality is lower because the accruals-based earnings are not backed by cash creation from operating activities.

Like the statement of cash flows, the income statement is a "flow" statement (sometimes referred to as a dynamic statement), as it shows the company's performance between two balance sheet dates. However, due to the accruals concept, net income is not the same as cash generated by the company. In addition, many financing and investing cash flows do not affect the income statement at the time of the transactions.

With respect to the balance sheet, operating activities relate (with a few exceptions) to the firm's current assets and current liabilities. Investing activities typically relate to the firm's noncurrent assets, and financing activities typically relate to the firm's noncurrent liabilities and equity.

Transactions for which the timing of revenue or expense recognition differs from the receipt or payment of cash are reflected in changes in balance sheet accounts. For example, when revenues (sales) exceed cash collections, the firm has sold items on credit, and accounts receivable (an asset) increase. The opposite occurs if customers repay more on their outstanding accounts than the firm extends in new credit: cash collections exceed revenues, and accounts receivable decrease. Similarly, when purchases from suppliers exceed cash payments, accounts payable (a liability) increase. When cash payments exceed purchases, accounts payable decrease.

It is helpful to understand how transactions affect each balance sheet account. For example, accounts receivable are increased by sales on credit and decreased by cash collections. We can summarize this relationship as follows:

Beginning accounts receivable	Opening balance sheet
+ Sales	Income statement
- <u>Cash collections</u>	Cash flow statement
= Ending accounts receivable	Closing balance sheet

Knowing three of the four variables, we can solve for the fourth. For example, if beginning accounts receivable are €10,000, ending accounts receivable are €15,000, and sales are €68,000, then cash collections must equal €63,000.

By rearranging this relationship, we can say that cash = sales – ending accounts receivable + beginning accounts receivable, or cash = sales – change in accounts receivable. Cash = €68,000 – (€15,000 – €10,000) = €63,000.

Revenue recognition standards affect when a sale is recorded in the income statement, but not the timing of cash flows from the customer. If a firm receives cash from a customer in advance of the sale being recorded in the income statement (unearned revenue), the cash received will appear in the cash flow statement at the time of payment, not when the sale is recorded. For example, if a company provided a service agreement to a customer, the cash received is reflected in the cash flow statement when the customer pays for the contract (normally in advance), but revenue recognized in the income statement is spread over the life of the contract.

Example: Balance sheet and income statement impacts on cash flow

Loftus Communications Limited provides equipment sales and installation of telecommunication equipment. Loftus also provides maintenance for systems installed under service contracts. Revenue from maintenance contracts is deferred and recognized over the life of the contract.

Selected Financial Statement Data: Loftus Communications Limited

Financial year	20X2	20X1
	£m	£m

Income statement extract

Revenues	2,000,000	1,800,000
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Balance sheet extract

Accounts receivable	900,000	500,000
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Unearned revenue (deferred) 1,000,000 300,000

Calculate the cash received from customers in 20X2.

Answer:

	20X2
	£m

Revenue 2,000,000

Increase in accounts receivable (400,000)

Increase in unearned revenue 700,000

Cash received from customers 2,300,000

The increase in accounts receivable represents credit extended to customers, which is a use of cash from the firm's perspective. The increase in unearned revenue represents services that customers have paid for in advance, a source of cash.

Understanding these interrelationships is not only useful in preparing the cash flow statement, but is also helpful in uncovering accounting shenanigans, as we will see in our reading on Financial Reporting Quality.

LOS 30.b: Describe the steps in the preparation of direct and indirect cash flow statements, including how cash flows can be computed using income statement and balance sheet data.

PROFESSOR'S NOTE

Throughout the discussion of the direct and indirect methods, remember the following points:

- The terms *cash flow from operations (CFO)*, *cash flow from operating activities*, and *operating cash flows (OCF)* are used interchangeably in the Level I CFA curriculum.
- Two methods can be used in the accounts to present CFO. CFO is presented differently under the direct and indirect methods, but the result is the same under both methods.
- Cash flow from investing (CFI) and cash flow from financing (CFF) are each presented the same way regardless of which method a firm chooses to present CFO.
- An increase in an asset account is a use of cash, and a decrease in an asset account is a source of cash.
- An increase in a liability account is a source of cash, and a decrease in a liability is a use of cash.
- Sources of cash are positive numbers (cash inflows), and uses of cash are negative numbers (cash outflows).

The Direct Method for Cash Flow From Operating Activities

The direct method of presenting a firm's statement of cash flows shows only cash payments and cash receipts over the period. The sum of these inflows and outflows is CFO. The direct method gives an analyst more information than the indirect method (which we will explain later in this reading). The analyst can see the actual amounts that went to each use of cash and that were received from each source of cash. This information can help the analyst to better understand the firm's performance over time and to forecast future cash flows.

The following are common components that appear on a statement of cash flows presented using the direct method:

- Cash collected from customers (typically the main component of CFO)
- Cash used in the production of goods and services (cash inputs)
- Cash operating expenses, such as salaries
- Cash paid for interest
- Cash paid for taxes

These are steps for the direct method:

Step
1: Start at the top of the income statement with revenue.

Step
2: Examine the balance sheet for any assets or liabilities (typically current) relating to the income statement item.

Step
3: Compute the change in the balance sheet asset or liability.

Adjust the income statement for the change in the balance sheet amount using the following rules:

Step
4:

- Subtract an increase in an asset (a use of cash), or add a decrease in an asset (a source of cash).
- Add an increase in a liability (a source of cash), or subtract a decrease in a liability (a use of cash).

For these rules to work consistently, we must *treat expense items as negative numbers* (e.g., cost of goods sold, wages and salaries) before we adjust them.

Step
5: After adjusting the income statement item for the change in the balance sheet asset or liability, move to the next item in the income statement.

Step
6: Ignore any noncash charges (e.g., depreciation, gains and losses on asset disposal). A noncash charge is any amount in the income statement that is due to accounting treatment rather than actual cash flows.

Step
7: Once all income statement items have been adjusted for accruals, total the amount to get CFO.

Example: CFO using the direct method

Use the following balance sheet and income statement to prepare operating cash flows using the direct method. The company produces its financial statements under U.S. GAAP.

Income Statement for 20X7	\$
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Sales	104,000
Expenses	
Cost of goods sold	40,000
Wages	5,000
Depreciation	7,000
Interest	<u>1,000</u>
Total expenses	<u>53,000</u>
Income from continuing operations	51,000
Gain from sale of land	10,000
Loss on disposals of PP&E	<u>2,000</u>
Pretax income	59,000

Provision for income taxes 20,000

Net income 39,000

Common dividends declared 8,500

Balance Sheets for 20X7 and 20X6

	20X7	20X6
Assets	\$	\$

Current assets

Cash 53,000 11,500

Accounts receivable 10,000 9,000

Inventory 5,000 7,000

Total current assets 68,000 27,500

Noncurrent assets

Land	35,000	40,000
Gross plant and equipment	69,000	60,000
Less: accumulated depreciation	<u>(12,000)</u>	<u>(9,000)</u>
Net plant and equipment	57,000	51,000
Goodwill	<u>10,000</u>	<u>10,000</u>
Total assets	170,000	128,500

Liabilities

Current liabilities

Accounts payable	9,000	5,000
Wages payable	4,500	8,000
Interest payable	3,500	3,000

Unearned revenue	6,000	2,000
Taxes payable	5,000	4,000
Dividends payable	<u>6,000</u>	<u>1,000</u>
Total current liabilities	34,000	23,000
Noncurrent liabilities		
Bonds payable	15,000	10,000
Deferred tax liability	<u>20,000</u>	<u>15,000</u>
Total liabilities	69,000	48,000
Stockholders' equity		
Common stock	15,000	20,000
Additional paid in capital	<u>25,000</u>	<u>30,000</u>

Contributed capital	40,000	50,000
Retained earnings	<u>61,000</u>	<u>30,500</u>
Total equity	<u>101,000</u>	<u>80,500</u>
Total liabilities and stockholders' equity	170,000	128,500

Answer:

Start at the top of the income statement and adjust each line for the change in balance sheet asset and liabilities that arise due to the accruals process.

Cash collected from customers	\$
Sales	104,000
Increase in accounts receivable	(1,000)
Increase in unearned revenue liability	<u>4,000</u>
Cash collected	107,000

To calculate cash paid to suppliers, adjust cost of goods sold for the change in inventory (the result is purchases for the period) and for the change in accounts payable.

Cash paid to suppliers	\$
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Cost of goods sold	(40,000)
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Decrease in inventory	<u>2,000</u>
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Purchases	(38,000)
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Increase in accounts payable	<u>4,000</u>
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Cash paid to suppliers	(34,000)
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All expenses, including cost of goods sold, must be treated as negative values if we wish to apply the rules for sources and uses of cash. Expenses in the income statement may be shown as either negative or positive values in the financial statements at the company's discretion. Users of the accounts are expected to know that revenue and gains increase net income, and that expenses and losses decrease it.

Wages paid	\$
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Wage expense	(5,000)
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Decrease in wages payable (3,500)

Cash paid to employees (8,500)

The next line in the income statement is depreciation. When using the direct method we ignore depreciation because it is not a cash flow.

Interest paid	\$
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Interest expense	(1,000)
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Increase in interest payable 500

Cash interest paid	(500)
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We also ignore the gain from sale of land and the loss on disposal of PP&E because these items relate to investing activities (CFI) and not operating activities (CFO). Later in this reading we will explain how to determine the related cash flows when we calculate CFI.

The last item in the income statement before net income is the tax expense. Here we must adjust not only for changes in the current liability for taxes payable, but also for any changes in deferred tax assets and deferred tax liabilities, which are typically noncurrent items. We will explain these further in our reading on Analysis of Income Taxes.

Cash paid to tax authorities	\$
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Tax expense (provision for income taxes) (20,000)

Increase in taxes payable liability	1,000
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Increase in deferred tax liability	<u>5,000</u>
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Cash paid for taxes	(14,000)
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Having arrived at the bottom of the income statement, we can sum the cash flows to compute CFO:

20X7 CFO

	\$	\$
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Cash collected from customers	107,000
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Cash paid to suppliers	34,000
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Wages paid to employees	8,500
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Cash interest paid	500
Cash paid to tax authorities	<u>14,000</u>
Total cash operating expenses	<u>(57,000)</u>
Operating cash flows	50,000

Module 30.2: Indirect Method CFO

SCHWESERNOTES - BOOK 2

Using the indirect method of presenting CFO, we begin with net income and adjust it for differences between accounting items and actual cash inflows and outflows.

Non-cash-based items in the income statement can be described as either noncash charges or working capital investment. For example, depreciation is a noncash charge as it is deducted in calculating net income, but it requires no cash outlay. Therefore, we must add depreciation (and amortization) back to net income for the period.

Another adjustment to net income on an indirect statement of cash flows is to subtract gains and add back losses on the disposal of assets. Proceeds from the sale of fixed assets are an investing cash flow. Because gains and losses relate to CFI activities, we need to remove them from net income to calculate CFO under the indirect method.

PROFESSOR'S NOTE

Candidates are often confused by why noncash charges and gains and losses are ignored when using the direct method, but adjusted for under the indirect method. The key to understanding this is to realize that these items have been included in arriving at net income, which is the starting point of our computation when we use the indirect method. That is, net income includes some non-cash and non-operating items that we have to unwind to arrive at CFO.

Under the indirect method, we also need to adjust net income for any change in balance sheet accounts, just as we do with the direct method. If, for example, accounts receivable went up during the period, we know that sales during the period were greater than the cash collected from customers. We need to reduce net income to reflect the fact that sales, rather than cash collected, were used in calculating net income.

The net change in a company's total operating assets and liabilities is known as its **investment in working capital**. We can think of CFO as net income adjusted for noncash charges and the investment in working capital:

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WC}_{\text{INV}}$$

Noncash charges are gains and losses that have passed through the income statement but are not cash flows. While we call them "charges," in practice they can either increase or decrease net income. Gains and losses on asset disposals are classic examples. **Typical Noncash Charges, Gains, and Losses** lists some of the typical noncash charges in company income statements.

Typical Noncash Charges, Gains, and Losses

Add Back Depreciation, depletion, and amortization

Loss on asset disposal

Asset impairments and write-downs

Losses on early retirement of debt

Amortization of bond discounts (under the amortized cost method)

Increases in deferred tax liabilities, decreases of deferred tax assets

Losses of equity accounted associates

Subtract Gains on asset disposals

Gains on early retirement of debt

Reversals of impairment and write-downs

Amortization of bond premiums (under the amortized cost method)

Decreases in deferred tax liabilities, increases in deferred tax assets

Working capital investment represents the investment in noncash working capital. This means we need to look at the change in current assets and liabilities that relate to the operating revenues and expenses and ignore any assets or liabilities that do not relate to operating items.

PROFESSOR'S NOTE

The definition of *working capital* is not the same in all parts of the Level I CFA curriculum. In our module covering ratios, we will see it defined as total current assets minus total current liabilities. Here and in Corporate Issuers, it is defined as operating assets minus operating liabilities; this is often referred to as noncash working capital.

In practice, when looking at the current assets, we ignore cash and any short-term investments (except trading securities, for which cash flows are treated as CFO). Dividends payable and any short-term interest-bearing debt instruments are also excluded, as they affect cash flow from financing (CFF), not CFO.

Adjust for changes in working capital accounts as follows:

Add back Decreases in current operating assets
 Increases in current operating liabilities

Subtract Increases in current operating assets
 Decreases in current operating liabilities

The steps in calculating CFO under the indirect method can be summarized as follows:

Step
1: Begin with net income.

Step
2: Add back all noncash charges to income (such as depreciation and amortization) and subtract all noncash components of revenue. Subtract gains or add losses that resulted from financing or investing cash flows (such as gains from sale of land).

Step
3: Adjust for working capital by adding or subtracting changes to balance sheet operating accounts as follows:

- Subtract increases in operating asset accounts (uses of cash), and add decreases (sources of cash).
- Add increases in operating liability accounts (sources of cash), and subtract decreases (uses of cash).

Example: Statement of cash flows using the indirect method

Use the balance sheet and income statement presented in the previous direct method example to prepare operating cash flows under the indirect method.

Answer:

Step 1: Start with net income of \$39,000.

Add back noncash charges:

Depreciation = \$7,000

Change in deferred tax liability = \$5,000

Step 2:

Loss on disposal of PP&E = \$2,000

Deduct noncash gains:

Gain from sale of land = \$10,000

Step 3: Subtract increases in receivables and inventories and add increases in payables.

	\$
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Net income	39,000
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Noncash charges

Depreciation	7,000
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Increase in deferred tax liability	5,000
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Loss on disposal of PP&E	2,000
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Gain from sale of land	<u>(10,000)</u>
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Subtotal	43,000
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Investment in working capital

Increase in receivables	(1,000)
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Decrease in inventories	2,000
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Increase in accounts payable	4,000
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Decrease in wages payable	(3,500)
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Increase in interest payable	500
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Increase in unearned revenue liability	4,000
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Increase in taxes payable	<u>1,000</u>
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Operating cash flows**50,000**

Alternatively, we can look at noncash working capital in aggregate rather than line by line.

	20X7	20X6
	\$	\$

Total current assets	68,000	27,500
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Less cash	(<u>53,000</u>)	(<u>11,500</u>)
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Operating assets	15,000	16,000
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Total current liabilities	34,000	23,000
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Less dividends payable	(<u>6,000</u>)	(<u>1,000</u>)
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Operating liabilities	28,000	22,000
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Non-cash working capital	(13,000)	(6,000)
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Investment in working capital (\$7,000)

We can then calculate CFO as $NI + NCC - WC_{INV}$:

$$CFO = 39,000 + 4,000 - (-7,000) = \$50,000$$

Our examples demonstrate that whether CFO is calculated using the direct or indirect method, we arrive at the same figure.

Both IFRS and U.S. GAAP encourage the use of a statement of cash flows in the direct format. Under U.S. GAAP, a statement of cash flows under the direct method must include footnote disclosure of the indirect method. Most companies, however, report cash flows using the indirect method, which requires no additional disclosure. Later in this reading we will illustrate the method an analyst can use to create a statement of cash flows in the direct method format when the company reports using the indirect method.

Module 30.3: Investing and Financing Cash Flows and IFRS/U.S. GAAP Differences

SCHWESERNOTES - BOOK 2

Cash flow from investing activities (CFI) consists of the cash inflows and outflows that result from acquiring or disposing of long-term assets and certain investments.

Cash flow from financing activities (CFF) consists of the cash inflows and outflows that result from transactions affecting a firm's capital structure, such as borrowing, repaying debt, and issuing or redeeming equity securities.

Examples of each cash flow classification, in accordance with U.S. GAAP, are presented in **U.S. GAAP Cash Flow Classifications**.

U.S. GAAP Cash Flow Classifications

Operating Activities

Inflows

Cash collected from customers

Outflows

Cash paid to employees and suppliers

Interest and dividends received

Cash paid for other expenses

Sale proceeds from trading securities

Acquisition of trading securities

Interest paid on debt or leases

Taxes paid

Investing Activities

Inflows

Outflows

Sale proceeds from PP&E and intangibles

Acquisition of PP&E and intangibles

Sale proceeds from debt and equity investments

Acquisition of debt and equity investments

Principal received from loans made to others

Loans made to others

Financing Activities

Inflows

Outflows

Principal amounts of debt issued

Principal paid on debt or leases

Proceeds from issuing stock

Payments to reacquire stock

Dividends paid to shareholders

Debt and equity investments (other than trading securities) and loans made to others are reported as investing cash outflows. However, under U.S. GAAP the income from these investments (interest and dividends received) is reported as operating cash inflows. Principal amounts borrowed are reported as financing cash inflows, but interest paid is reported as an operating cash outflow. Finally, dividends paid to the firm's shareholders are financing cash outflows. Later in this reading we will see that IFRS offers companies more discretion than U.S. GAAP with regard to how they may classify these cash flows.

PROFESSOR'S NOTE

Don't confuse dividends received and dividends paid. Under U.S. GAAP, dividends received are operating cash flows, and dividends paid are financing cash flows.

Example: Computing CFI

Returning to the financial statements we used in our CFO examples, we will use the following information to compute CFI:

	20X7	20X6
Balance sheet extract	\$	\$

Land	35,000	40,000
Gross PP&E	69,000	60,000
Accumulated depreciation	<u>(12,000)</u>	<u>(9,000)</u>
Net PP&E	57,000	51,000

Income statement extract

Depreciation	7,000
Gain on sale of land	10,000
Loss on disposal of PP&E	2,000

A footnote disclosure reveals that the company purchased PP&E for \$25,000 during 20X7.

Answer:

CFI will comprise additions to PP&E and the disposal proceeds from the sale of PP&E and land.

The footnote disclosure shows acquisitions of PP&E were \$25,000; however, gross PP&E has only increased by \$9,000. This, coupled with the disposal loss in the income statement, indicates that the company must have disposed of PP&E during 20X7.

Step 1: Compute cost of disposed PP&E \$

Beginning gross PP&E	60,000
Acquisitions gross cost	25,000
Disposals gross cost	(X)
Ending gross PP&E	69,000

The value (X) we need to calculate is the cost of the assets that have been disposed of; that is, the gross book value the company recorded when it first acquired these assets. By rearranging the reconciliation, the gross cost of the disposed asset is computed as follows:

beginning PP&E + acquisitions – ending gross PP&E = disposals gross cost

disposals gross cost = \$60,000 + \$25,000 – \$69,000 = \$16,000

Next, we can see that accumulated depreciation has increased by \$3,000 when the 20X7 depreciation expense is \$7,000. The difference is also due to the asset disposal, because when a company disposes of an asset, it removes the accumulated depreciation on that asset from the total.

Step 2: Compute accumulated depreciation on disposed PP&E \$

Beginning accumulated depreciation	9,000
Depreciation expense	7,000
Accumulated depreciation on disposed PP&E	(X).
Ending accumulated depreciation	12,000

Accumulated depreciation on disposed PP&E = beginning accumulated depreciation + depreciation expense
– ending accumulated depreciation = \$9,000 + \$7,000 – \$12,000 = \$4,000.

Step 3: Compute carrying value of PP&E disposal \$

Gross cost	16,000
Accumulated depreciation	(<u>4,000</u>).
Carrying value before disposal	12,000

PROFESSOR'S NOTE

Steps 1–3 can be combined for a quicker result:

Shortcut approach	\$
Beginning carrying value	51,000
Depreciation expense	(7,000)
Additions to PP&E	25,000
Carrying value of assets disposed (X)	
Ending carrying value	57,000

Carrying value of assets disposed = beginning carrying value – depreciation expense + additions to PP&E – ending carrying value = \$51,000 – \$7,000 + \$25,000 – \$57,000 = \$12,000

When an asset is disposed of, the carrying value is removed from the balance sheet and netted against proceeds received from the sale, and any difference is reported as an accounting gain or loss in the income statement. A gain results if the proceeds from the sale exceed the carrying value, and a loss results if the proceeds are less than the carrying value. The only elements of the disposal that are cash flows are the disposal proceeds.

Step 4: Compute disposal proceeds	\$
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Disposal proceeds	X
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Carrying value removed from balance sheet	<u>(12,000)</u>
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Disposal gain/(loss) in income statement	(2,000)
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The disposal proceeds on the sale of PP&E must have been \$10,000.

We carry out similar calculations for the disposal of land. This will be simpler because land is not depreciated, so the carrying value is its gross cost. Because the footnotes do not mention any acquisitions of land, the carrying value of the land disposed of is the change in carrying value reported on the balance sheet.

Carrying value of disposed land = \$40,000 – \$35,000 = \$5,000

Compute disposal proceeds	\$
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Disposal proceeds	X
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Carrying value removed from balance sheet	<u>(5,000)</u>
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Disposal gain/(loss) in income statement	10,000
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The disposal proceeds on the sale of land must have been \$15,000.

Finally we can combine these results to determine cash flow from investing:

CFI = -cash paid for PP&E acquisitions + disposal proceeds

$$\text{CFI} = -\$25,000 + \$10,000 + \$15,000 = \$0$$

The cost of new assets acquired was, by coincidence, perfectly matched by the proceeds from asset disposals.

Example: Computing cash flow from financing (CFF)

Returning to the financial statements from our CFO examples, we will need the following information to compute CFF:

	20X7	20X6
Balance sheet extract	\$	\$

Current liabilities

Dividends payable	6,000	1,000
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Noncurrent liabilities

Bonds payable	15,000	10,000
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Stockholders' equity

Common stock	15,000	20,000
Additional paid-in capital	<u>25,000</u>	<u>30,000</u>
Contributed capital	40,000	50,000
Retained earnings	61,000	30,500

Other financial statement data

Net income	39,000
Dividend declared	8,500

A footnote disclosure reveals that the bonds outstanding had been issued at face value (par).

Answer:

We can begin by determining cash flows from issuing or repaying bonds:

\$

Beginning bonds payable 10,000

Net principal flows X/(X)

Ending bonds payable 15,000

net principal flow = ending bonds payable – beginning bonds payable = \$15,000 – \$10,000 = \$5,000
(inflow)

Next we can determine cash flows from issuing or redeeming equity shares. **Contributed capital** is the sum of common stock at par and additional paid-in capital, and reflects the price at which the company issued shares.

Reconciliation of equity contributed capital \$

Beginning contributed capital 50,000

Net proceeds from repurchases and issuance X/(X)

Ending contributed capital 40,000

net proceeds from buyback and issuance = ending contributed capital – beginning contributed capital =
\$50,000 – \$40,000 = –\$10,000 (use of cash, or a net share repurchase of \$10,000)

Next we determine cash dividends paid, which can be a two-step process:

Step 1: Calculate dividend declared if it is not given.

In this example, this dividend declared of \$8,500 was given. If it had not been, we could compute it based on the change in retained earnings:

Reconciliation of retained earnings		\$
Beginning retained earnings		30,500
Net income		39,000
Dividend declared		(X)
Ending retained earnings		61,000

$$\begin{aligned}\text{dividends declared} &= \text{beginning retained earnings} + \text{net income} - \text{ending retained earnings} = \\ & \$30,500 + \$39,000 - \$61,000 = \$8,500\end{aligned}$$

Step 2: Adjust the dividend declared for changes in the dividends payable liability. Simply because it is declared does not mean it has been paid. While the dividend declared does not pass through the income statement, it does reduce retained earnings; therefore, we treat it as a negative value. We then apply the increase/decrease rules for balance sheet liabilities:

Dividends paid		\$
Dividend declared		(8,500)

Increase in dividend payable liability 5,000

Cash dividend paid (3,500)

Computation of CFF	\$
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Net principal on bonds 5,000

Net proceeds from share repurchase and issuance (10,000)

Cash dividends (3,500)

Cash flow from financing (8,500)

Having computed CFO, CFI, and CFF, we can complete the cash flow statement.

Total cash flow	\$
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Cash flow from operations 50,000

Cash flow from investing 0

Cash flow from financing (8,500)

Total cash flow 41,500

20X6 balance sheet cash 11,500

20X7 balance sheet cash 53,000

The total cash flow of \$41,500 is equal to the increase in cash. The difference between beginning cash and ending cash should be used as a check figure to ensure that the total cash flow calculation is correct.

PROFESSOR'S NOTE

The Level I CFA curriculum makes a few simplifications in the calculation of CFF.

For bonds issued at a premiums or discounts, the difference relative to par is amortized over the life of the bond. Amortization of premiums and discounts are not cash flows:

coupon + amortized discount – amortized premium = interest expense

With *amortization of a discount bond*, amortization will increase the interest expense and carrying value.

With *amortization of a premium bond*, amortization will decrease the interest expense and carrying value.

The curriculum mentions the noncash element affecting interest expense (see **Typical Noncash Charges, Gains, and Losses**), but does not mention the impact on the balance sheet carrying value.

The Level I curriculum sidesteps such complications by assuming that bonds are issued at par value in computations of CFF, which means there are no premiums or discounts to amortize.

A second simplification in the Level I curriculum is to assume that stock buybacks are transacted at the same price that the shares were initially issued for. In practice, this is unlikely, and if the buyback price differs from issue price, adjustments to retained earnings are made.

LOS 30.c: Demonstrate the conversion of cash flows from the indirect to direct method.

The only difference between the indirect and direct methods of presentation is in the cash flow from operations (CFO) section. CFO under the direct method can be computed using a combination of the income statement and a statement of cash flows prepared under the indirect method.

PROFESSOR'S NOTE

The Level I CFA curriculum describes a three-step method for converting from indirect cash flow statements to direct cash flow statements. Provided that you understood the direct and indirect methods presented earlier, you will find little that you do not already know (although it is presented slightly differently).

Here is the three-step process:

Step 1: Aggregate all revenues and gains and all expenses and losses.

Step 2: Remove all noncash charges and disaggregate the remaining items.

Step 3: Convert from accruals to cash flows by adjusting for the change in working capital.

Example: Conversion from indirect to direct CFO

Using the same data as the direct and indirect examples, the three stages are as follows:

\$

Step 1:

Total revenues and gains	114,000
Total expenses and losses	<u>75,000</u>
Net income	39,000

Step 2:

Revenues less noncash charges	
(\$114,000 – \$10,000)	104,000

\$10,000 = gain on asset disposal

Expenses less noncash charges	
(\$75,000 – \$7,000 – \$2,000 – \$5,000)	61,000

\$7,000 = depreciation

\$2,000 = loss on disposal of PP&E

\$5,000 = change in deferred tax liability

Cost of goods sold	40,000
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Wages	5,000
Interest	1,000
Tax payable	<u>15,000</u>
(Tax provision – increase in DTL)	
Total	61,000

Step 3:

Cash collected from customers	107,000
$\$104,000 - \$1,000 + \$4,000$	
Cash paid to suppliers	(34,000)
$-\$40,000 + \$2,000 + \$4,000$	
Cash paid to employees	(8,500)
$-\$5,000 - \$3,500$	

Cash interest paid	(500)
–\$1,000 + \$500	
Cash paid to tax authorities*	(14,000)
–\$15,000 + \$1,000	
Net cash flow from operating activities	50,000

* Note that the computation of cash paid looks slightly different from the direct method example. This is because the change in deferred tax is treated as a noncash charge and adjusted for in Step 2.

LOS 30.d: Contrast cash flow statements prepared under International Financial Reporting Standards (IFRS) and US generally accepted accounting principles (US GAAP).

Recall that under U.S. GAAP, dividends paid to the firm's shareholders are reported as financing activities, while interest paid is reported in operating activities. Interest received and dividends received from investments are also reported as operating activities.

International Financial Reporting Standards (IFRS) allow more flexibility in the classification of cash flows. Under IFRS, interest and dividends received may be classified as either operating *or* investing cash inflows. Dividends paid to the company's shareholders and interest paid on the company's debt may be classified as either operating *or* financing cash outflows.

Another important difference relates to income taxes paid. Under U.S. GAAP, all taxes paid are reported as operating cash outflows—even taxes related to investing and financing transactions. Under IFRS, income taxes are reported as operating cash outflows unless they are associated with an investing or financing transaction.

For example, consider a company that sells land that was held for investment for \$1 million. Income taxes on the sale total \$160,000. Under U.S. GAAP, the firm reports an inflow of cash from investing activities of \$1 million and an outflow of cash from operating activities of \$160,000. Under IFRS, the firm can report a net inflow of \$840,000 from investing activities.

The differences between U.S. GAAP and IFRS are summarized in **Differences Between U.S. GAAP and IFRS**.

Differences Between U.S. GAAP and IFRS

	U.S. GAAP	IFRS
Interest received	CFO	CFO or CFI
Interest paid	CFO	CFO or CFF
Dividends received	CFO	CFO or CFI
Dividends paid	CFF	CFO or CFF
Bank overdraft	Treated as balance sheet debt	Treated as balance sheet cash
Taxes	CFO	May be split between CFO, CFI, and CFF according to the nature of the transaction that caused tax to become payable

Presentation of
CFO

Direct preferred, but indirect allowed; a
reconciliation of net income to CFO must be
disclosed if using direct method

Direct preferred, but indirect allowed

Reading 30: Key Concepts

SCHWESERNOTES - BOOK 2

LOS 30.a

Cash flow from operations is not the same as earnings because of the accruals process. To calculate CFO, balance sheet operating assets and liabilities are used to adjust income statement revenues and expenses to cash flows.

Cash flows can be computed as the income statement figure – increase in related operating assets + decreases in related operating assets + increases in related operating liabilities – decrease in related operating liabilities.

Operating activities typically relate to the firm's current assets and current liabilities. Investing activities typically relate to noncurrent assets. Financing activities typically relate to noncurrent liabilities and equity.

LOS 30.b

Under the direct method of presenting CFO, each line item of the accruals-based income statement is adjusted to get cash receipts or cash payments. The main advantage of the direct method is that it presents clearly the firm's operating cash receipts and payments.

Under the indirect method of presenting CFO, net income is adjusted for transactions that affect net income but do not affect operating cash flow, such as depreciation and gains or losses on asset sales, and for changes in balance sheet items. The main advantage of the indirect method is that it focuses on the differences between net income and operating cash flow and gives the user of the accounts an indication of earnings quality.

CFI is calculated by determining the changes in asset accounts that result from investing activities. The cash flow from selling an asset is its book value plus any gain on the sale (or minus any loss on the sale).

CFF is the sum of net cash flows from creditors (new borrowings minus principal repaid) and net cash flows from shareholders (new equity issued minus share repurchases minus cash dividends paid).

LOS 30.c

An indirect cash flow statement can be converted to a direct cash flow statement by adjusting each income statement account for changes in associated balance sheet accounts and by eliminating noncash and nonoperating items (i.e., applying the direct method).

LOS 30.d

Differences in cash flow classifications between U.S. GAAP and IFRS:

U.S. GAAP		IFRS
Interest received	CFO	CFO or CFI
Interest paid	CFO	CFO or CFF
Dividends received	CFO	CFO or CFI
Dividends paid	CFF	CFO or CFF

Bank overdraft Treated as balance sheet debt

Treated as balance sheet cash

Taxes CFO

May be split between CFO, CFI, and CFF
according to the nature of the transaction that
caused tax to become payable

Presentation of
CFO Direct preferred but indirect allowed; a
reconciliation of net income to CFO must be
disclosed if using the direct method

Direct preferred but indirect allowed
